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YEAR/SEM/SEC: Y3/S5/B**

**MINI PROJECT  
LIBRARY INFORMATION SYSTEM TYPES**

**OF COMMANDS:**

**DDL[DATA DEFINITION LANGUAGE]**

**1.CREATE:**

**BOOKSS:**

```
SQL> select*from BOOKSS;
```

| AUTHOR   | ISBN  | TITLE    | EDITION | CATEGORY | PRICE |
|----------|-------|----------|---------|----------|-------|
| silber   | 10101 | database | 7       | CSE      | 25000 |
| samantha | 20202 | ds       | 5       | CSE      | 35000 |
| albert   | 24170 | os       | 6       | IT       | 10000 |

**READERS:**

```
SQL> select*from READERS;
```

| USER_ID | EMAIL   | ADDRESS | PHONE_NO | NAME  |
|---------|---------|---------|----------|-------|
| 1010    | t@gmail | pondy   | 1317     | thanu |
| 2003    | r@gmail | chennai | 2431     | ramya |
| 2002    | d@gmail | kerala  | 3113     | dina  |

### PUBLISHERSS:

```
SQL> select*from PUBLISHERSS;
```

| PUBLISHER_ | YEAR_OF_PUBLICATION | NAME   |
|------------|---------------------|--------|
| -----      | -----               | -----  |
| 1724       | 2004                | karthi |
| 1106       | 1995                | abhi   |
| 6060       | 2000                | janu   |

### AUTHENTICATION:

```
SQL> select*from AUTHENTICATION;
```

| LOGIN_ID | PASSWORD |
|----------|----------|
| -----    | -----    |
| hema     | hema24   |
| sana     | sana17   |
| ashwin   | ashwin04 |

### REPORTS:

```
SQL> select*from REPORTS;
```

| USER_ID | REG_NO   | BOOK_NO | ISSUE_DATE |
|---------|----------|---------|------------|
| 2024    | 21td0065 | 1317    | 24/10/2023 |
| 2526    | 22td0010 | 2431    | 10/01/2023 |
| 8691    | 19td0015 | 3544    | 06/11/2023 |

**STAFF:**

```
SQL> select*from STAFF;
```

| NAME      | STAFF_ID |
|-----------|----------|
| hemalatha | hema     |
| pugal     | riya     |
| mohan     | jaivanth |

**2.ALTER:**

```
SQL> alter table TEACHERS add(area char(10));
```

```
Table altered.
```

```
SQL> select*from TEACHERS;
```

| NAME      | STAFF_ID | AREA |
|-----------|----------|------|
| hemalatha | hema     |      |
| pugal     | riya     |      |
| mohan     | jaivanth |      |
| varshini  | vasu     |      |

**3.DROP:**

```
SQL> drop table REPORTS;

Table dropped.

SQL> select*from REPORTS;
select*from REPORTS
      *
ERROR at line 1:
ORA-00942: table or view does not exist
```

#### **4.TRUNCATE:**

```
SQL> truncate table PUBLISHERSS;

Table truncated.

SQL> select*from PUBLISHERSS;

no rows selected
```

#### **5.RENAME:**

```
SQL> rename STAFF to TEACHERS;
```

```
Table renamed.
```

```
SQL> select*from TEACHERS;
```

| NAME      | STAFF_ID |
|-----------|----------|
| hemalatha | hema     |
| pugal     | riya     |
| mohan     | jaivanth |
| varshini  | vasu     |

## DML[DATA MANIPULATION LANGUAGE]

### 1.SELECT:

```
SQL> select*from BOOKSS;
```

| AUTHOR   | ISBN  | TITLE    | EDITION | CATEGORY | PRICE |
|----------|-------|----------|---------|----------|-------|
| silber   | 10101 | database | 7       | CSE      | 25000 |
| samantha | 20202 | ds       | 5       | CSE      | 35000 |
| albert   | 24170 | os       | 6       | IT       | 10000 |

### 2.INSERT:

```
SQL> Insert into STAFF(name,staff_id)values('varshini','vasu');

1 row created.

SQL> select*from STAFF;
```

| NAME      | STAFF_ID |
|-----------|----------|
| hemalatha | hema     |
| pugal     | riya     |
| mohan     | jaivanth |
| varshini  | vasu     |

### 3.UPDATE:

```
SQL> update BOOKSS set price='15000' where author='albert';

1 row updated.

SQL> select*from BOOKSS;
```

| AUTHOR   | ISBN  | TITLE    | EDITION | CATEGORY | PRICE |
|----------|-------|----------|---------|----------|-------|
| silber   | 10101 | database | 7       | CSE      | 25000 |
| samantha | 20202 | ds       | 5       | CSE      | 35000 |
| albert   | 24170 | os       | 6       | IT       | 15000 |

### 4.DELETE:

```
SQL> delete from PUBLISHERSS where name='karthi';

1 row deleted.

SQL> select*from PUBLISHERSS;
```

| PUBLISHER_ | YEAR_OF_PUBLICATION | NAME |
|------------|---------------------|------|
| 1106       | 1995                | abhi |
| 6060       | 2000                | janu |

## TRANSACTION CONTROL COMMMANDS:

### 1.DESCRPTION:

```
SQL> desc REPORTS;
Name                                     Null?      Type
-----
USER_ID                                VARCHAR2(10)
REG_NO                                VARCHAR2(20)
BOOK_NO                                VARCHAR2(10)
ISSUE_DATE                             VARCHAR2(15)
```

### 2.COMMIT:

```
SQL> commit;

Commit complete.

SQL> select*from PUBLISHERSS;

PUBLISHER_ YEAR_OF_PUBLICATION  NAME
-----
1106        1995                  abhi
6060        2000                  janu
```

### 3.SAVEPOINT:

```
SQL> savepoint sana;
```

Savepoint created.

```
SQL> select*from AUTHENTICATION;
```

| LOGIN_ID | PASSWORD |
|----------|----------|
| hema     | hema24   |
| sana     | sana17   |
| ashwin   | ashwin04 |

#### 4.ROLLBACK:

```
SQL> delete from BOOKSS where author='samantha';
```

1 row deleted.

```
SQL> select*from BOOKSS;
```

| AUTHOR | ISBN  | TITLE    | EDITION | CATEGORY | PRICE |
|--------|-------|----------|---------|----------|-------|
| silber | 10101 | database | 7       | CSE      | 25000 |
| albert | 24170 | os       | 6       | IT       | 15000 |

```
SQL> roll back;
```

Rollback complete.

```
SQL> select*from BOOKSS;
```

| AUTHOR   | ISBN  | TITLE    | EDITION | CATEGORY | PRICE |
|----------|-------|----------|---------|----------|-------|
| silber   | 10101 | database | 7       | CSE      | 25000 |
| samantha | 20202 | ds       | 5       | CSE      | 35000 |
| albert   | 24170 | os       | 6       | IT       | 15000 |

## ARITHMETIC OPERATORS



#### 1.MAX:

```
SQL> select max(price)from BOOKSS;

MAX(PRICE)
-----
35000
```

#### 2.MIN:

```
SQL> select min(price)from BOOKSS;

MIN(PRICE)
-----
15000
```

#### 3.COUNT:

```
SQL> select count(name)from STAFF;

COUNT(NAME)
-----
4
```

#### 4.AVERAGE:

```
SQL> select avg(price)from BOOKSS;

AVG(PRICE)
-----
      25000
```

#### **5.SUM:**

```
SQL> select sum(price)from BOOKSS;

SUM(PRICE)
-----
      75000
```

#### **SET OPERATORS:**

##### **UNION:**

```
SQL> select user_id from READERS union select user_id from REPORTS;

USER_ID
-----
1010
2002
2003
2024
2526
8691

6 rows selected.
```

##### **UNION ALL:**

```
SQL> select login_id from AUTHENTICATION union all select staff_id from STAFF;

LOGIN_ID
-----
hema
sana
ashwin
hema
riya
jaivanth
vasu

7 rows selected.
```

## MINUS:

```
SQL> select login_id from AUTHENTICATION minus select staff_id from STAFF;

LOGIN_ID
-----
ashwin
sana
```

## INTERSECT:

```
SQL> select login_id from AUTHENTICATION intersect select staff_id from STAFF;

LOGIN_ID
-----
hema
```

## DATA QUERY LANGUAGE:

### 1.SELECT:

```
SQL> select distinct title from BOOKSS;

TITLE
-----
os
database
ds
```

### 2.ORDER BY CLAUSE:

```
SQL> select email from READERS where name='dina' order by email;

EMAIL
-----
d@gmail

SQL> select * from PUBLISHERSS order by year_of_publication desc,publisher_id asc;

PUBLISHER_ YEAR_OF_PUBLICATION  NAME
-----
6060      2000      janu
1106      1995      abhi
```

### 3.GROUP BY CLAUSE:

```
SQL> select name from STAFF group by name;

NAME
-----
varshini
pugal
mohan
hemalatha

SQL> select author,avg(price) as averageprice from BOOKSS group by author;

AUTHOR      AVERAGEPRICE
-----
albert      15000
silber      25000
samantha    35000
```